VIDEO GAME SALES ANALYSIS PROJECT

PROJECT OVERVIEW

The global video game industry is one of the largest and fastest-growing entertainment sectors in the world. Understanding what drives sales performance, user engagement, and market trends is essential for publishers, developers, and business analysts.

This project focuses on analyzing video game sales data combined with engagement and rating metrics to uncover patterns related to:

- Genre performance
- Platform dominance
- Regional sales behavior
- User engagement impact
- Rating influence on revenue
- Market growth over time

The analysis is performed using a complete data pipeline approach, from raw data cleaning to dashboard visualization.

The main objectives of this project are:

1. To clean and structure raw gaming datasets for analysis.
2. To design a relational SQL database for efficient querying.
3. To analyze global and regional sales trends.
4. To study the relationship between ratings, engagement, and revenue.
5. To identify high-performing genres and platforms.
6. To build interactive Power BI dashboards for business decision-making.

Datasets Used

1- games.csv (Metadata + Engagement Data)
Contains:
• Game title
• Release date
• Developer/Team
• Rating

• Plays
• Wishlist count
• Backlogs
• Genre

2- vgsales.csv (Sales Data)
Contains:
• Game name
• Platform
• Year
• Publisher
• NA Sales
• EU Sales
• JP Sales
• Other Sales
• Global Sales

Note:
All sales values are measured in MILLIONS.

Project Workflow

The project was executed in three major stages:

Stage 1 – Data Cleaning & Exploration (Python in VS Code)

- Cleaned missing values
- Standardized column names
- Converted data types
- Removed duplicates
- Performed exploratory data analysis (EDA)

Stage 2 – Database Structuring (MySQL)

- Created structured relational tables
- Designed merged dataset (Sales + Engagement + Ratings)
- Wrote analytical SQL queries
- Validated totals and aggregates

Stage 3 – Data Visualization (Power BI)

- Built interactive dashboards
- Designed KPIs and calculated measures
- Analyzed trends across time, genre, region, and platform
- Extracted business insights

Business Questions Addressed

This project answers key strategic questions such as:

- Which genres generate the most global sales?
- Which platforms dominate the market?
- Which region contributes the highest revenue?
- Does higher user rating lead to more sales?
- Do wishlists and engagement influence profitability?
- How has the gaming market evolved over time?
- What are the strongest genre-platform combinations?

Project Significance

This project demonstrates a complete real-world analytics workflow:

- Raw data preprocessing
- Database design and query optimization
- Statistical exploration
- Business intelligence dashboard development
- Data-driven insight generation

It showcases practical skills in:

- Data Cleaning
- SQL Querying
- Relational Database Design
- Data Visualization
- Analytical Thinking
- Business Interpretation

Sample Cleaning Code

```python
games["Rating"] = games["Rating"].fillna(games["Rating"].median())
```

```python
games["Team"] = games["Team"].fillna("unknown")
games["Summary"] = games["Summary"].fillna("unknown")
```

```python
sales = sales.dropna(subset=["Year"])
sales["Year"] = sales["Year"].astype(int)
```

```python
sales["Publisher"] = sales["Publisher"].fillna("unknown")
```

Outcome of Data Cleaning

- Removed inconsistent records
- Standardized data types
- Ensured reliable aggregation
- Prepared dataset for SQL modeling and Power BI dashboards

EDA 1: Regional Sales Performance by Platform

```python
sales.groupby("Platform")[["NA_Sales", "EU_Sales", "JP_Sales"]] \
    .sum() \
    .sort_values("NA_Sales", ascending=False) \
    .head(10)
```

Python

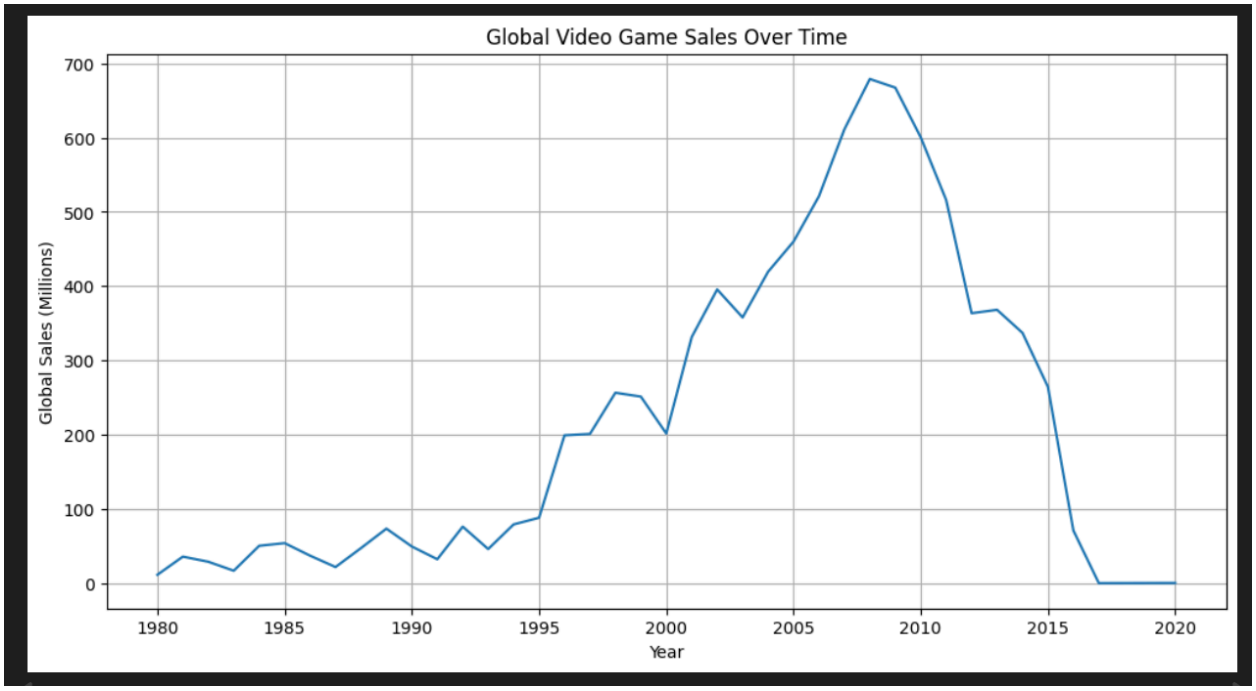
Platform	NA_Sales	EU_Sales	JP_Sales
X360	594.33	278.01	12.30
PS2	572.92	332.63	137.54
Wii	497.37	264.35	68.28
PS3	388.90	340.47	79.21
DS	388.56	194.07	175.02
PS	334.71	212.38	139.78
GBA	184.12	74.59	46.56
XB	182.06	59.65	1.38
N64	138.91	41.03	33.76
GC	131.94	38.32	21.34

Explanation:

- Aggregated total NA, EU, and JP sales for each gaming platform using groupby and sum.
- Identified top 10 platforms based on highest North American sales.
- Revealed regional market differences, showing Xbox dominance in NA and stronger Nintendo/PlayStation performance in Japan and Europe.

EDA 2: Global Video Game Sales Trend Over Time

```python
plt.figure(figsize=(12,6))
plt.plot(yearly_sales.index, yearly_sales.values)
plt.title("Global Video Game Sales Over Time")
plt.xlabel("Year")
plt.ylabel("Global Sales (Millions)")
plt.grid(True)
plt.show()
```



Explanation:

- The line chart shows the yearly trend of total global video game sales (in millions) from 1980 to 2020.
- Sales gradually increased from the 1980s, experienced rapid growth between 2000–2008, and peaked around 2008–2009.
- After 2010, there is a noticeable decline in sales, indicating a shift in the gaming industry (rise of digital distribution, mobile gaming, and changing market dynamics).

SQL DATABASE DESIGN & QUERIES

We created a structured relational database.

Database Creation

```
CREATE DATABASE video_game_sales;
USE video_game_sales;
```

Merged Dataset Creation

```sql
CREATE TABLE merged_data AS
SELECT
    s.name,
    s.platform,
    s.year,
    s.genre,
    s.publisher,
    s.na_sales,
    s.eu_sales,
    s.jp_sales,
    s.other_sales,
    s.global_sales,
    g.rating,
    g.plays,
    g.wishlist,
    g.backlogs
FROM sales s
LEFT JOIN games_clean g
ON s.name = g.title;
```

Important SQL Queries Used in Analysis

Total Global Sales

SELECT SUM(global_sales) FROM merged_data;

Result Grid | Filter Rows: | Export: | Wrap C

	SUM(global_sales)
▶	8820.359997089952

Purpose:
Calculate overall market size.

Top Genres by Sales

SELECT genre, SUM(global_sales) AS total_sales

```
FROM merged_data
GROUP BY genre
ORDER BY total_sales DESC;
```

Purpose:
Identify most profitable game categories.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

genre	total_sales
Action	1722.8799964878708
Sports	1309.2399954497814
Shooter	1026.1999989431351
Role-Playing	923.8400027602911
Platform	829.1500045042485
Misc	797.619999974966
Racing	726.7699981294572
Fighting	444.0499999728054
Simulation	390.16000044718385
Puzzle	242.22000078856945
Adventure	234.7999997511506
Strategy	173.4299998804927

Regional Sales Comparison

```
SELECT
   SUM(na_sales) AS total_na,
   SUM(eu_sales) AS total_eu,
   SUM(jp_sales) AS total_jp
FROM merged_data;
```

Purpose:
Determine strongest global region.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

total_na	total_eu	total_jp
4333.430002428591	2409.119998589158	1284.299998562783

Sales Trend Over Time

```
SELECT year, SUM(global_sales)
FROM merged_data
GROUP BY year
ORDER BY year;
```

Purpose: Analyze industry growth cycle.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

year	SUM(global_sales)
1980	11.379999861121178
1981	35.77000015974045
1982	28.860000044107437
1983	16.790000066161156
1984	50.35999968647957
1985	53.940001683309674
1986	37.07000032067299
1987	21.740000121295452
1988	47.220000518485904
1989	73.44999986514449
1990	49.39000025391579
1991	32.2300001680851
1992	76.16000100970268
1993	45.980000203475356
1994	79.17000025883317
1995	88.10999972373247
1996	199.15000126510859

High Rated Platforms

```
SELECT platform, AVG(rating)
FROM merged_data
GROUP BY platform
ORDER BY AVG(rating) DESC;
```

Purpose:
Find platforms producing better-rated games.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

platform	AVG(rating)
SAT	4.05000
SNES	3.84091
GC	3.71935
DS	3.70000
PS	3.70000
PS2	3.69467
XB	3.69259
3DS	3.68182
DC	3.67500
GB	3.66364
N64	3.63158
GBA	3.61333
Wii	3.57000
PC	3.56807
PSV	3.55185
WiiU	3.55000
PS3	3.53873

POWER BI DASHBOARDS

We created 3 professional dashboards.

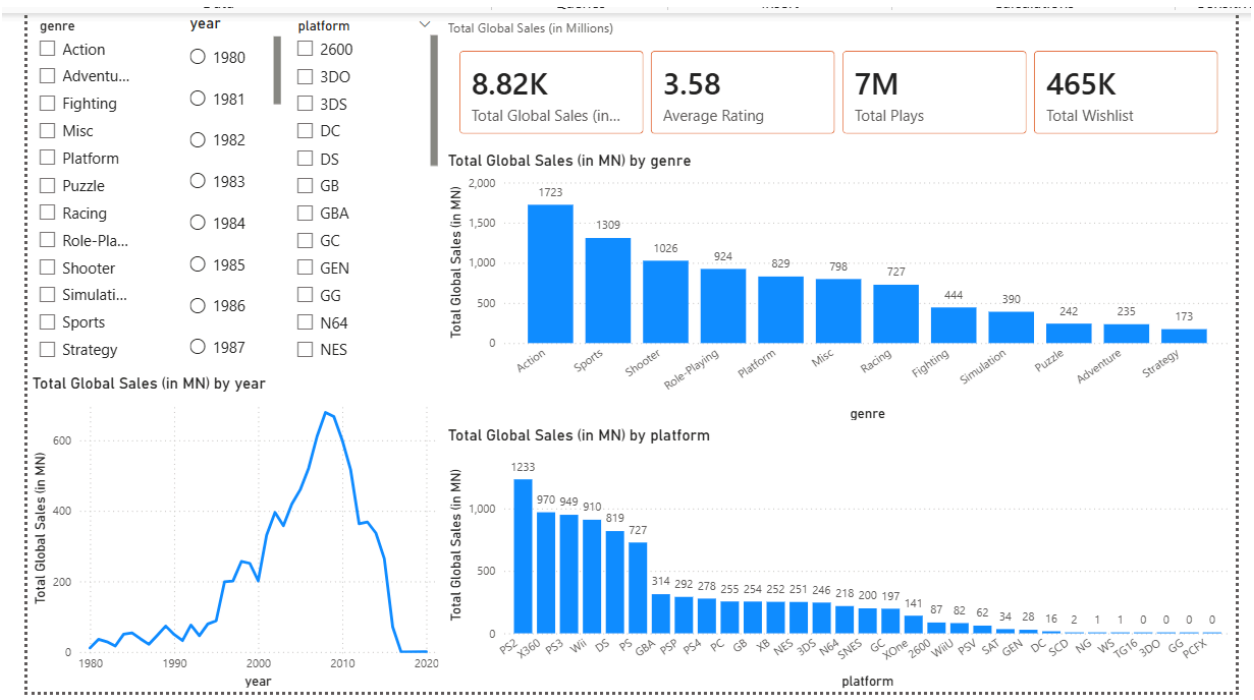
DASHBOARD 1 – EXECUTIVE OVERVIEW

KPIs:

- Total Global Sales
- Average Rating
- Total Plays
- Total Wishlist

Charts:

- Global Sales by Genre (Bar Chart)
- Global Sales by Platform (Bar Chart)
- Global Sales Trend by Year (Line Chart)



Sales Performance Summary

This dashboard provides a high-level summary of global video game sales performance across genres, platforms, and years. It is designed to give decision-makers a quick understanding of market size, trends, and top-performing segments.

KPI Cards (Top Section)

1. Total Global Sales (8.82K)
   - Represents total global sales in millions.
   - 8.82K = 8,820 million units (or $8.82 billion if interpreted in revenue terms).
   - Shows overall market size.

2. Average Rating (3.58)
   - Average user rating across all games.
   - Helps understand overall user satisfaction level.

3. Total Plays (7M)
   - Total number of recorded user plays.
   - Indicates engagement level of users across all games.

4. Total Wishlist (465K)
   - Total number of times games were added to wishlist.
   - Reflects user interest and demand potential.

Chart 1: Total Global Sales (in MN) by Genre

Purpose:
To identify which game genres generate the highest global sales.

Key Insights:
- Action genre leads the market with the highest sales.
- Sports and Shooter genres follow closely.
- Strategy and Adventure genres contribute comparatively less.

Conclusion:
Action-oriented games dominate the global market. Developers targeting mass revenue should focus on Action, Sports, and Shooter genres.

Chart 2: Total Global Sales (in MN) by Year

Purpose:
To analyze sales trends over time.

Key Insights:
- Slow growth from 1980 to mid-1990s.
- Strong growth phase between 1998–2008.
- Peak around 2008–2009 (~680M).
- Sharp decline after 2012.

Conclusion:
The industry experienced a golden era between 2000–2010. Post-2012 decline may reflect dataset limitations or industry transition toward digital/mobile platforms.

Chart 3: Total Global Sales (in MN) by Platform

Purpose:
To identify best-selling gaming platforms.

Key Insights:
- PS2 leads with highest total global sales.
- X360, PS3, and Wii are strong performers.
- Older platforms like NES, N64, and handheld consoles show moderate impact.
- Some platforms show near-zero sales (possibly due to limited data).

Conclusion:
Console generations such as PS2, X360, PS3, and Wii drove the majority of historical sales. Platform strategy significantly impacts revenue performance.

Overall Business Insights from Dashboard 1

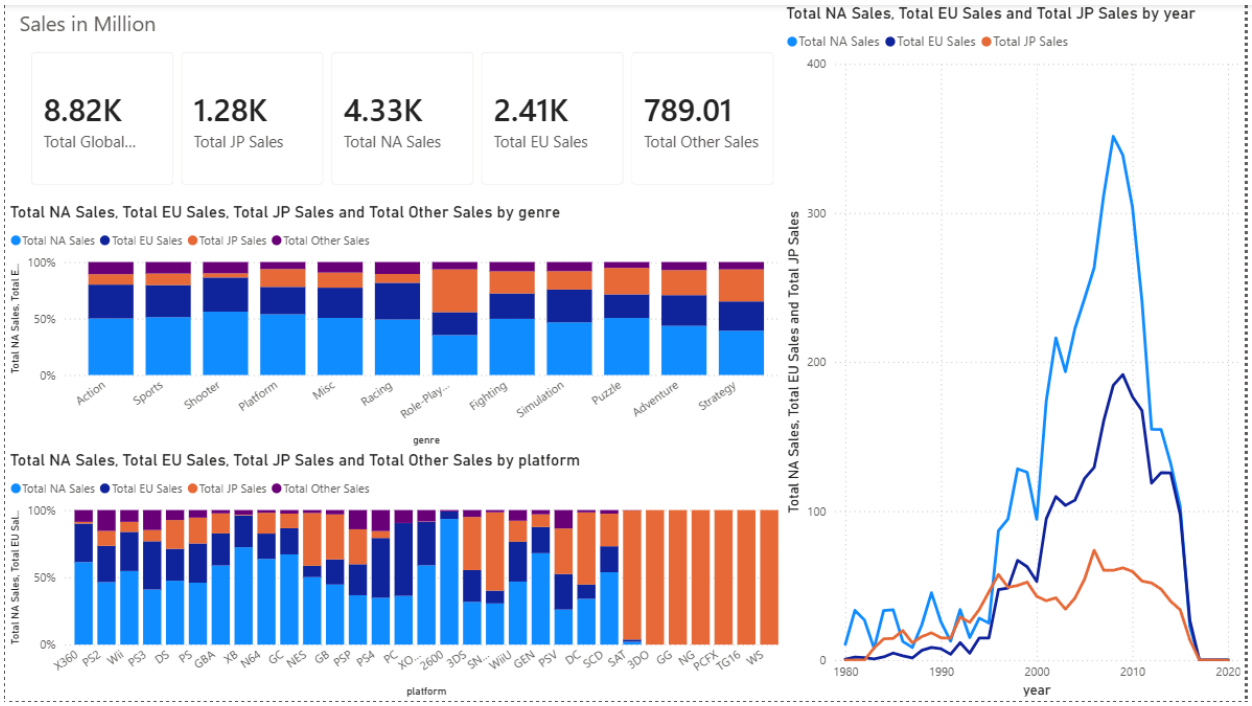
1. The video game market generated over 8.8 billion (in millions units) globally.
2. Action and Sports genres dominate revenue generation.
3. The peak sales period was around 2008–2009.
4. Console platforms such as PS2 and X360 were major revenue drivers.
5. Engagement (7M plays) is strong, indicating an active user base.

Why This Dashboard Matters

- Provides executive-level overview of performance.
- Identifies profitable genres and platforms.
- Highlights market evolution over time.
- Supports strategic decisions in game development and publishing.

This dashboard answers:
- Which genres generate the most sales?
- Which platforms dominate the market?
- How has the industry evolved over time?
- What is the overall market size and engagement level?

DASHBOARD 2 – PLATFORM & MARKET ANALYSIS



This dashboard focuses on regional sales performance across genres, platforms, and time.
It helps us understand which regions dominate the market and how regional preferences differ.

KPI CARDS (Sales in Millions)

- Total Global Sales: 8.82K (≈ 8.82 Billion)
- Total NA Sales: 4.33K
- Total EU Sales: 2.41K
- Total JP Sales: 1.28K
- Total Other Sales: 789.01

Purpose:
These KPIs provide a high-level summary of total revenue contribution by each region.

Key Insight:
North America contributes the largest share of global video game sales, followed by Europe and Japan.

Regional Sales by Genre (Stacked Column Chart)

Chart: Total NA, EU, JP, and Other Sales by Genre

Purpose:
To compare how different regions contribute to each genre's total sales.

Key Observations:
- Action and Sports genres generate strong sales across all regions.
- Role-Playing games show relatively stronger performance in Japan compared to other genres.
- North America consistently dominates across most genres.
- Strategy and Puzzle genres contribute smaller overall revenue.

Conclusion:
Regional genre preferences vary slightly, but NA remains the dominant market overall.

Regional Sales by Platform (Stacked Column Chart)

Chart: Total NA, EU, JP, and Other Sales by Platform

Purpose:
To analyze how each platform performs regionally.

Key Observations:
- X360 and PS2 perform extremely well in North America.
- PlayStation platforms (PS2, PS3, PS4) show strong presence in Europe.
- Nintendo platforms (DS, Wii) show noticeable performance in Japan.
- Some platforms are region-dominant (e.g., Japanese platforms performing better in JP region).

Conclusion:
Platform success varies by geography.
Console manufacturers have stronger performance in specific regions.

Regional Sales Trend Over Time (Line Chart)

Chart: NA, EU, JP Sales by Year

Purpose:
To analyze how regional sales evolved over time.

Key Observations:

- Global sales peak between 2008–2010.
- North America shows the highest peak.
- Europe follows a similar trend pattern.
- Japan remains comparatively lower but stable.
- Sharp decline after 2015 indicates market shift (digital distribution & incomplete recent data).

Conclusion:
The video game industry experienced major growth during the 2000–2010 era,
followed by market decline in physical sales data.

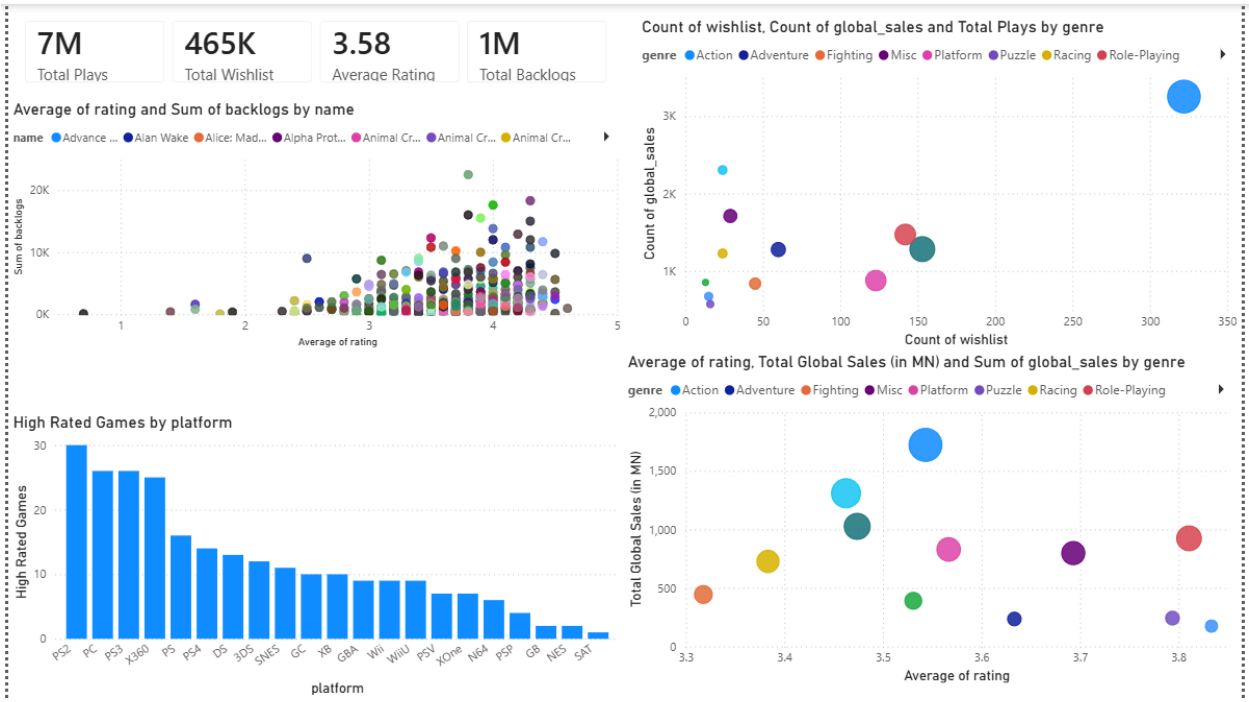
Overall Dashboard Insights

- North America is the dominant revenue region.
- Action and Sports are universally strong genres.
- Platform popularity differs by geography.
- The industry peaked around 2008–2010.
- Market behavior varies regionally but follows similar growth patterns.

This dashboard successfully answers:

- Which region generates the most sales?
- How do regional sales compare by genre?
- How do platforms perform regionally?
- How has regional performance changed over time?

DASHBOARD 3 – ENGAGEMENT & RATING ANALYSIS



Objective:
This dashboard analyzes how user engagement metrics (plays, wishlist, backlogs) and ratings relate to global sales performance.
It focuses on behavioral insights rather than only financial performance.

KPI Section (Top Cards)

1. Total Plays - 7M
   - Represents total user engagement across all games.
   - Indicates how frequently games are being played.
   - Shows overall popularity beyond just sales numbers.

2. Total Wishlist - 465K
   - Reflects future demand and user interest.
   - Helps measure pre-purchase or long-term interest.

3. Average Rating - 3.58
   - Overall average user rating across all games.
   - Used to analyze correlation between ratings and sales.

4. Total Backlogs - 1M
   - Represents games users own but haven't completed.

• Indicates strong engagement but possible content depth.

Chart 1: Average Rating vs Sum of Backlogs (by Game Name)

What it shows:
• X-axis: Average Rating
• Y-axis: Total Backlogs
• Each dot represents a game.

Insights:
• Most highly rated games (3.5 – 4.5 range) also show higher backlog counts.
• Popular games tend to accumulate more unfinished play sessions.
• Indicates that higher engagement may increase backlog volume.

Conclusion:
Higher-rated games generally maintain stronger long-term engagement.

Chart 2: Wishlist vs Global Sales vs Total Plays (by Genre)

What it shows:
• X-axis: Count of Wishlist
• Y-axis: Count of Global Sales
• Bubble size: Total Plays
• Each bubble represents a genre.

Insights:
• Action genre has the largest bubble → highest plays + strong sales.
• Some genres show high wishlist but moderate sales.
• Engagement (plays) appears strongly linked to sales performance.

Conclusion:
Genres with higher engagement levels tend to generate stronger sales.

Chart 3: High Rated Games by Platform

What it shows:
• Bar chart of number of games with high ratings (e.g., rating ≥ 4) by platform.

Insights:
• PS2, PC, and PS3 platforms have the highest number of highly rated games.
• Older legacy platforms still maintain strong quality perception.
• Platforms with more quality titles build stronger brand trust.

Conclusion:
Platform success is not just about sales volume but also quality output.

Chart 4: Average Rating vs Global Sales (by Genre)

What it shows:
• X-axis: Average Rating
• Y-axis: Total Global Sales (in millions)
• Bubble size: Total Sales volume.

Insights:
• Action genre leads in both rating and sales.
• Some genres have high ratings but lower sales (e.g., Puzzle, Strategy).
• Rating positively influences sales, but genre popularity plays a larger role.

Conclusion:
Ratings contribute to sales performance, but genre demand has stronger impact.

Overall Insights from Dashboard 3

1. Engagement (plays, wishlist, backlog) correlates positively with sales.
2. Higher-rated games generally perform better commercially.
3. Action and Adventure dominate both engagement and revenue.
4. Some niche genres maintain strong ratings but lower commercial performance.
5. Platform quality output influences long-term brand perception.

Final Takeaway

This dashboard moves beyond revenue analysis and focuses on behavioral patterns.
It demonstrates that:

- User engagement drives commercial success.
- Ratings support sales growth.
- Popular genres dominate both user interest and revenue.

Final Conclusion

The Video Game Sales & Engagement Analysis project successfully transformed raw, unstructured datasets into a structured, insight-driven business intelligence solution using Python, SQL, and Power BI.

Through systematic data cleaning in Python, relational structuring in SQL, and interactive visualization in Power BI, we were able to uncover meaningful patterns across sales, genres, platforms, regions, ratings, and user engagement metrics.

Key Achievements

1. Data Engineering
   - Cleaned missing values and standardized data using Python (Pandas).
   - Converted raw CSV files into structured SQL tables.
   - Created a merged dataset combining Sales + Ratings + Engagement metrics.
   - Ensured data integrity and consistency across tools.

2. Exploratory Data Analysis (EDA)
   - Identified top-performing platforms (PS2, X360, PS3).
   - Found Action and Sports as the highest revenue-generating genres.
   - Observed a strong industry growth period between 2000–2010.
   - Identified regional dominance of North America in total sales.

3. Business Intelligence Insights (Power BI)
   - Built 3 interactive dashboards:
     - Executive Overview
     - Platform & Market Analysis
     - Engagement & Rating Analysis
   - Measured correlation between ratings and global sales.
   - Analyzed wishlist and backlog behavior as engagement indicators.
   - Compared regional sales distribution by genre and platform.
   - Evaluated platform quality through high-rated game distribution.

Major Business Insights

• North America generates the highest share of global sales.
• Action genre dominates both revenue and engagement.
• Higher-rated games generally perform better commercially.
• Strong user engagement (plays, wishlist) positively correlates with sales.
• The industry experienced peak growth around 2008–2010.
• Platform success depends on both sales volume and quality output.

Overall Impact

This project demonstrates how data analytics can:

• Transform raw datasets into actionable insights.
• Support strategic decision-making for publishers and developers.
• Identify profitable genres and platforms.
• Analyze user behavior to predict commercial success.
• Provide a 360-degree market view through interactive dashboards.

Final Statement

By integrating data cleaning, SQL structuring, exploratory analysis, and business intelligence visualization, this project delivers a complete end-to-end data analytics workflow.

It not only answers key business questions but also showcases strong practical skills in:

- Python (Data Cleaning & EDA)
- SQL (Data Modeling & Querying)
- Power BI (Dashboarding & Insight Generation)
- Data Interpretation & Business Analysis

This project reflects a real-world analytics pipeline and demonstrates the ability to convert data into meaningful business intelligence.

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